Research Methods for IT Professionals (REM475S/REM474S/RME477S)

Advanced Diploma – Information Technology Department

Subject Guide, 2020

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Welcome and Introduction to the Subject

In our contemporary lives we are constantly bombarded with opinions that are presented as fact. Often these 'opinions' are said to be supported by research. However, research can mean different things in different environments and frequently the basis of the research supporting these 'opinions' have been shown to be invalid or untrue. An essential part of living in our media-dominated society is the ability to understand how valid, reliable and trustworthy scientific research is produced. Such understanding and awareness will enable you to become more critical about the nature of the facts, knowledge and information encountered on a daily basis.

*In this year-long course the nature, purpose and characteristics of scientific and academic research will be explored. As part of this exploration we will discuss other ways in which 'research' is understood and how it is different from the more systematic and structured scientific research process. The focus of the course is on the field of **RESEARCH METHODOLOGY** which is concerned with and sets out the rules, procedures, principles and values that guide scientific and academic researchers and enable them to produce reliable, credible and valid knowledge. The subject will help you understand and apply the different principles of scientific research and provide you with basic guidelines so that you are able to undertake your own small scale research project as a IT Professional.*

Teaching and Learning Goals

Learning Outcomes

1. Propose, design and conduct a small-scale research investigation, located within the IT professional context using basic questionnaire design
2. Produce and create various communication documents (written, oral, visual) that demonstrate appropriate academic conventions and practices
3. Exemplify appropriate ethical standards/values associated with academic research
4. Describe, illustrate and examine at an introductory level the different methodologies, methods, processes, procedures and techniques associated with legitimate academic/scientific research

Teaching Objectives

- Introduce students to the knowledge, principles, techniques, methods and philosophies associated with scientific/academic research as it applies to the IT disciplinary and professional contexts
- Guide student through the design, planning, implementation (conducting) and reporting of a small-scale research investigation, using basic questionnaire design and located in an IT industry/professional context
- Explore and discuss the different academic conventions and practices associated with reporting research findings
- Provide practice opportunities to produce different academic documents associated with a research undertaking e.g. written reports, oral presentations
- Introduce students to the principles, values and behaviours of an academic researcher that shown understanding of and adheres to the basic ethical standards and prescriptions

Content Topics

SEMESTER ONE (Feb – June)	
Term 1	*Resources*
In this topic area we will address the key question of "what defines scientific research". We will start this exploration by looking at what characterises scientific research and how it differs from other forms of day-to-day research which many of us already undertake. We will also discuss the different kinds/type of scientific research and how different researchers describe the kind of research they undertake. Part of this discussion will be the identification of the different procedures and activities associated with scientific research, why it is important to conduct research in an ethical manner and how researchers typically communicate their research work to the broader scientific community through academic writing. **SPECIFIC LESSON TOPICS** • Nature of academic/scientific research • Processes/activities/ Procedures associated with research • Different types of research (basic/applied/exploratory /descriptive • Being an ethical researcher • Writing like an academic researcher	• Chapter 1: Thinking in Print (Booth et al, 2016 – The Craft of Research) • Chapter 2: The Research Process (Kumar, 2014 – Research Methodology) • The Ethics of research (plagiarism) (Booth et al, 2016 – The Craft of Research) • Chapter 4: Ethical issues in social science research (Terre Blanche et al, 2006 – Research in Practice) • Online resources and notes
Term 2	*Resources*
The main focus of our discussions during this topic area, will be to expand our understanding of the very specific ways in which scientific research is structured and organised. You will be introduced to the 'research process' and the different steps, procedures and activities researchers undertake when conducting a scientific research study. These procedures and activities form the cornerstone of a scientific research study and helps to ensure that the research obtains valid results. You will also be introduced to a basic model of critical reflection. You will be assisted to use this model to reflect on your own journey to become an ethical researcher. **SPECIFIC LESSON TOPICS** • The research process • The research proposal • Research aims/objectives • Data collection (methods, participants, sampling) • Data collection activities • Critical reflection	• Chapter 3: Research Design (Terre Blanche et al, 2006 – Research in Practice) • Chapter 1: All at sea but learning to swim (Blaxter et al, 2006 – How to research 3rd Ed) • Chapter 2: The research process • Chapter 7: The research design • Chapter 12: Selecting a sample (Kumar, 2014 – Research Methodology) • Chapter 10: Sampling (Maree, 2016 – First steps in research 2nd Ed) • Chapter 6: Collecting data (Blaxter et al, 2006 – How to research 3rd Ed)
SEMESTER TWO (Aug– Dec)	
Term 3	*Resources*
In this topic area attention will be placed on data collection. Our discussion will include different methods of collecting information on a research topic. You will also be introduced to the specific ways of ensuring that the information you collect is of high quality and valid. We will undertake a detailed exploration of how to design valid questionnaires. You will be	• Chapter 9: Surveys and questionnaires (Maree, 2016 First steps in research 2nd Ed) • General introduction to the design of questionnaires

given multiple opportunities to design and implement your own questionnaire. We will also discuss the different ethical considerations a researcher has to take on board to ensure that their research does not cause any harm.	(Burgess, 2001, University of Leeds) • Chapter 14: Considering ethical issues (Kumar, 2014 – Research Methodology)
SPECIFIC LESSON TOPICS • Data collection instruments (questionnaire design) • Validity • Ethics	
Term 4	*Resources*
Our final topic area deals with the analysis of questionnaire data collected. We will consider ways to describe and summarise the information collected from the questionnaires you design and administered, in the form of 'Findings'. Practical opportunities will also be provided to consider the ways in which you can communicate your research process and findings in the form for a written research report.	• Chapter 16: Displaying data (Kumar, 2014 – Research Methodology) • Chapter 15: Communicating evidence visually • Chapter 17: Revising style (Booth et al, 2016 – The craft of research)
SPECIFIC LESSON TOPICS • Data analysis • Describing findings • Writing your research	

Classroom Sessions

During our classroom sessions you will be required to become an **active learner**; this means you have to do things; like take notes, respond to questions, discuss with your peers, do written activities, work on a computer, read, work with your peers in groups, speak to your peers and your lecturer, think and concentrate. You will have to complete the preparatory reading and/or writing tasks before coming to class – this will ensure you are able to engage with the subject content in a deep and intelligent way, and so gain the most from your very short time with your lecturer.

- In order to be successful in your studies, in addition to working with your lecturer during classroom periods, you will also have to devote a significant amount of time working on the subject content and learning materials outside classroom time. You should spend at least four (4) hours per week with either
    - Self-study and independent research
    - Library and internet learning resources
- You are strongly encouraged to consult with your lecturers to discuss your learning in this subject.

Blackboard will be used as a Learning Management Tool. Please check Blackboard regularly for both subject content and general communication about classes and assess. All official communications about this subject will take place via Blackboard.

Suggested Allocations for Teaching and Learning per Week

LEARNING ACTIVITY	PERCENTAGE (%) OF YOUR LEARNING TIME
Lectures (face to face contact time with your lecturer in a designated classroom)	30% (2 hours) over 24 weeks
Tutorials: individual groups of 15 or less	Not applicable
Practical workplace experience (experiential learning/work-based learning etc)	Not applicable
Independent self-study of standard texts and references (books, journal, online resources)	60% (4 hours) over 30 weeks (includes assessment periods)
Independent self-study of specially prepared materials (case studies, using online tutorials etc)	
Independent self-study, time needed to prepare for assessment (e.g. assignments, exams, projects)	
Student learning groups (working with your peers for assignments**)**	10% (30mins) over 30 weeks (includes assessment periods)

Overview of Weekly Lesson Topics

Week	Dates	Lesson Topic or Learning Activity
		TERM 1
1	3 - 7 Feb	Welcome and orientation to subject (What do researchers 'look like and do?')
2	9 - 13 Feb	What is research? Nature of academic research
3	16 - 20 Feb	What is research? Characteristics and types of research
4	23 - 27 Feb	Research activities – processes, procedures and activities associated with research
5	2 - 6 Mar	Conducting ethical research. Becoming an ethical researcher
6	9 - 13 Mar	Academic writing. Writing like a researcher
7	16 - 20 Mar	EXAM PREP

TERM 2

WEEK	Dates	Lesson Topic or Learning Activity	Readings
1	1-5 June	Orientation and Assessment	
2	8-12 June	Planning your research – the research process	Blaxter et al (2006) Chapter 1 Kumar (2011) Chapter 2
3	15-19 June (16 Jun: Youth Day)	The research proposal – Describing your research plan	Research Proposal Guideline Online resources
4	22-26 Jun	The research proposal – Research Topic, Research Problem and the Literature	Kumar (2011) Chapters 3&4
5	29 Jun – 3July	Collecting data: Types of data	Blaxter et al (2006) Chapter 6
6	6-10 July	Collecting data: Sampling	Maree (2016) Chapter 10 Kumar (2014) Chapter 12
7	13 – 17 July	Assessment - Research Proposal Support	
8	20 – 24 July		
9	27 -31 July		

TERM 3

1	17/18 August	ASSESSMENT FEEDBACK. Research Projects
2	24/25 August	Questionnaire design
3	31 Aug/1 Sept	Research Project: Research objectives and questions
4	7/8 September	Research Project: Questionnaire pilot & feedback
5	14/15 September	Research report
6	21/22 September	Research report
7	28/29 September	Assessment Support
		END of TERM 9 October

TERM 4

1	19/20 October	ASSESSMENT FEEDBACK
2	26/27 October	Research Project: Data analysis
3	2/3 November	Research Project: Reporting your findings
4	9/10 November	Revision and Assignment support
5	16/17 November	Revision and Assignment support
6		EXAM PERIOD:
7		EXAM PERIOD
8		**END of TERM 18 Dec 2020**

Assessment Strategies

Overview

Semester	Types of Assessment	Nature of Assessment	Assessment Weight	NEW
ONE	1. Close-book exam	Individual	10	SAME
	2. Written research proposal	PAIRS	30	SAME but different weight
	~~3. Reflective piece (1)~~	~~Individual~~	~~10~~	REMOVED
TWO	3. Research Report (part 1 & 2)	PAIRS	Part 1 = 20 Part 2 = 20	Part 1 = 20 Part 2 = 20
	4. Research Presentation	PAIRS	10	SAME
	5. Reflective piece	Individual	10	SAME

Demonstrating your Learning

In this subject you will be asked to undertake different kinds of tasks and activities so that you can demonstrate that you have learnt and understood the content, topics and skills taught in the classroom.

- The assessments in this subject will ask you to demonstrate what you KNOW, show that you can APPLY or USE this knowledge in practice, while also express your BELIEFS and VALUES about scientific research.
- It is very important to understand that your understanding, interpretation and application of the various concepts, guidelines, standards and approaches are more important that your ability to memorise facts.
- Look carefully at the distribution of marks for each assignment – most of the assignments require you to apply your knowledge and understanding to written research projects or reports.
- You will receive assessment/learning feedback after each assignment your complete. Feedback will be presented either individually or more generally to the whole class. These sessions are crucial for helping you establish what you are done well and the areas that need more improvement.

Academic writing is seen as central to your learning in this subject. You have to take reading and writing activities very seriously. You will also be provided with multiple opportunities to improve your academic writing, especially for the assessment requirements in this subject.

Assessment Types, Criteria and link to Outcomes

<table>
<tr><th>No</th><th>Semester</th><th colspan="2">ASSESSMENT TYPE and DESCRIPTION</th></tr>
<tr><td rowspan="3">1.</td><td rowspan="2">ONE</td><td colspan="2">CLOSED BOOK TEST (10%)</td></tr>
<tr><td colspan="2">What are the characteristics of academic/scientific research?
Distinctions between academic and other forms of research
Research process
Activities associated with research
Different types of research
What is ethical research practice</td></tr>
<tr><td colspan="2">ASSESSMENT CRITERIA
Test Memo</td><td>LEARNING OUTCOMES
4</td></tr>
</table>

<table>
<tr><th>No</th><th>Semester</th><th colspan="2">ASSESSMENT TYPE and DESCRIPTION</th></tr>
<tr><td rowspan="3">2</td><td>ONE</td><td colspan="2">WRITE A RESEARCH PROPOSAL (30%)</td></tr>
<tr><td colspan="3">Using the research proposal guideline, and working in pairs, produce a written research proposal that outlines the research design of a small-scale applied research investigation located in the IT professional context
3-5 pages long</td></tr>
<tr><td colspan="2">ASSESSMENT CRITERIA
Academic writing & style
Suitable reference list
Appropriate decision-making w.r.t research design and selection of appropriate research methods
Coherence and alignment of research elements</td><td>LEARNING OUTCOMES
1, 2, 4</td></tr>
</table>

<table>
<tr><th>No</th><th>Semester</th><th colspan="2">ASSESSMENT TYPE and DESCRIPTION</th></tr>
<tr><td rowspan="3">3.</td><td rowspan="2">TWO
(Term 3)</td><td colspan="2">RESEARCH REPORT (PART ONE) (20%)</td></tr>
<tr><td colspan="2">Working in pairs produce a written draft research report of the small-scale applied research investigation you are currently undertaking. The draft should report on your work-in-progress. Your report should describe/outline the following activities you have undertaken (1500-2000 words or roughly 3-5 pages)
Title
Introduction
Literature Review
Problem Statement
Aims, objectives and research questions
Data collection
Ethical Considerations
Reference list</td></tr>
<tr><td colspan="2">ASSESSMENT CRITERIA
Academic writing & style
Topic discussion, quality of literature review
Appropriate delineation of problem statement, aims and objectives
Description of research questions
Questionnaire Design
Ethical awareness and considerations</td><td>LEARNING OUTCOMES
1,2,3,4</td></tr>
</table>

<table>
<tr><th>No</th><th>Semester</th><th colspan="2">ASSESSMENT TYPE and DESCRIPTION</th></tr>
<tr><td rowspan="3">3.</td><td rowspan="2">TWO
(Term 4)</td><td colspan="2">RESEARCH REPORT (PART TWO) (20%)</td></tr>
<tr><td colspan="2">Working in pairs produce a written draft research report of the small-scale applied research investigation you are currently undertaking. The final report should provide a comprehensive description of your completed study. The report should be 7-10 pages long (max of 5000 words). It should include the following headings
Title
Introduction
Literature Review
Problem Statement
Aims, objectives and research questions
Research Design (data collection, data analysis, ethical considerations)
Findings
Discussion and Recommendations
Conclusion
Reference list</td></tr>
<tr><td colspan="2">ASSESSMENT CRITERIA
Academic writing & style
Appropriate delineation of problem statement, aims and objectives
Decision making wrt research design & research Methods
Analysis and interpretation of data
Coherence & alignment of research elements</td><td>LEARNING OUTCOMES
1,2,3,4</td></tr>
</table>

<table>
<tr><th>No</th><th>Semester</th><th colspan="2">ASSESSMENT TYPE and DESCRIPTION</th></tr>
<tr><td rowspan="3">4.</td><td rowspan="2">TWO
(Term 3)</td><td colspan="2">RESEARCH PRESENTATION (10%)</td></tr>
<tr><td colspan="2">Working in pairs, prepare a short 5-7 minute presentation that reports on your work-in-progress activities linked to your small-scale research investigation</td></tr>
<tr><td colspan="2">ASSESSMENT CRITERIA
Presentation skills
Appropriate delineation of objectives, research questions
Coherence in research design
Questionnaire design
Analysis and interpretation of data</td><td>LEARNING OUTCOMES
1,2,4</td></tr>
<tr><th>No</th><th>Semester</th><th colspan="2">ASSESSMENT TYPE and DESCRIPTION</th></tr>
<tr><td rowspan="3">5.</td><td rowspan="2">TWO
(Term 4)</td><td colspan="2">WRITTEN CRITICAL REFLECTION (10%)</td></tr>
<tr><td colspan="2">Write a short (max 600 words) reflective review that describes your activities as an ethical researcher undertaking your first research study</td></tr>
<tr><td colspan="2">ASSESSMENT CRITERIA
Academic writing & style
Application of critical reflection model
Quality of reflection</td><td>LEARNING OUTCOMES
2,3</td></tr>
</table>

Assessment Guidelines for the Subject

- All assignments you have to complete will be communicated to you via an assignment brief. Your lecturer will either give you a print version of the assignment or explain it in class. This brief will contain everything you need to know about what you have to do for the assignment task. All briefs will also be uploaded on Blackboard.
- Briefs will also include the **criteria** that will be used to mark or judge your learning. All the assignments for your subject will be discussed in detail in class.
- All assignments must be submitted in HARD COPY.
- Deadlines for all assignments will be STRICTLY enforced. Late submission penalties of -5% per day will be applied for the first four (4) days, after which the assignment will no longer be accepted.
- Marked assignments/test will be returned to students for comment within a 14-day period. Students will have five (5) days to raise any concerns or objections with the mark received (please refer to CPUT Assessment Rules).
- Students are entitled to follow the CPUT procedures to raise queries or objections regarding assessments via the department or faculty (see CPUT Assessment Rules).

Learning Resources

Main Textbooks:

1) **Kumar, Ranji, 2011** – Research Methodology: A step by step guide for beginners, 3rd Edition (available as downloadable PDF). 2014 4th Edition available for purchase

Supplementary Textbooks:

1. **Booth et al, 2016 – The Craft of Research, University of Chicago Press: Chicago**
   ISBN-13: 9780226239736
   a. Chapter 1: Thinking in Print
   b. The Ethics of research (plagiarism)
   c. Chapter 17: Revising style
2. **Terre Blanche, M and Durrheim, K and Painter, D. (editors) 2006 – Research in Practice 2nd Edition, UCT Press: Cape Town**
   ISBN: 9781919713694
   a. Chapter 3: Research Design (Durrheim, Kevin)
   b. Chapter 4: Ethical issues in social science research (Wassenaar, Douglas)
3. **Blaxter et al, 2006 – How to research 3rd Ed, Open University Press: Maidenhead**
   ISBN: 033521746X
   a. Chapter 1: All at sea but learning to swim
   b. Chapter 6: Collecting data

4. **Maree, 2016 – First steps in research 2nd Ed Chapter, Van Schaik: Pretoria**
   ISBN: 9780627033698
   a. Chapter 9: Surveys and questionnaires (Maree, Kobus and Pietersen, Jacques)
   b. Chapter 10: Sampling (Maree, Kobus and Pietersen, Jacques)

Administrative Information

Assessment Guidelines for the Subject

- All assignments you have to complete will be communicated to you via an assignment brief. Your lecturer will either give you a print version of the assignment or explain it in class. This brief will contain everything you need to know about what you have to do for the assignment task. All briefs will also be uploaded on Blackboard.
- Briefs will also include the **criteria** that will be used to mark or judge your learning. All the assignments for your subject will be discussed in detail in class.
- All assignments must be submitted in HARD COPY.
- Deadlines for all assignments will be STRICTLY enforced. Late submission penalties of -5% per day will be applied for the first four (4) days, after which the assignment will no longer be accepted.
- Marked assignments/test will be returned to students for comment within a 14-day period. Students will have five (5) days to raise any concerns or objections with the mark received (please refer to CPUT Assessment Rules)

Assessment Dates

Semester	Types of Assessment	Assessment Weight	Proposed Dates	NEW Dates
ONE	~~1. Close-book exam~~	~~10~~	~~March 2020~~	~~COMPLETE~~
	2. Written research proposal	30	May 2020	17 July
TWO	3. Research Report (part 1 & 2)	Part 1 = 20 Part 2 = 20	Mid-August 2020 Early-October 2020	**Part 1** 14 Sept **Part 2** - 2 Nov
	4. Research Presentation	10	Mid-October 2020	21 Sept
	5. Reflective piece	10	Mid-October 2020	13 Nov

CPUT Assessment Rules

1) See pages 18 – 20 for Assessment Rules in the General Handbook: https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20_Regulations_2018.pdf
2) Assessment and Graduation Centre: http://www.cput.ac.za/services/agc

Other Useful Institutional Resources

Research Support Guide (http://libguides.library.cput.ac.za/research)
Campus Health Clinics (http://www.cput.ac.za/students/life/clinics)
Student Counselling (http://www.cput.ac.za/students/life/counselling)
Financial Aid and Funding (http://www.cput.ac.za/study/funding)
NSFAS (http://www.nsfas.org.za/content/)
Student Representatives Council (http://www.cput.ac.za/students/about/src)
Student Development (http://www.cput.ac.za/students/life/development)

CPUT GRADUATE ATTRIBUTES

Graduate attributes are the qualities, skills and understandings that the CPUT community agrees you will develop as a student should during their time with the institution. These **attributes** include but go beyond the disciplinary or subject expertise or technical knowledge that form part of the different subjects you will complete as part of your diploma.

1. Technological capability and foresight

A CPUT graduate will recognise that society, technology and science are intertwined, so that technology and science have the capacity to effect changes in society. Furthermore, CPUT graduates will recognise that science and technology should be used for the overall benefit of society even though its effects may also sometimes be harmful. graduates will also recognise that scientific knowledge and their related technologies will need to be transformed/adapted to complex and changing circumstances. CPUT graduates would thus need to take a critical and reflective stance on how technology is used and for what ends, including issues of environmental awareness and sustainability, and to act accordingly.

2. Resilience and problem solving capability

A CPUT graduate will recognise the complexity of problem solving in society (including technological problem-solving) and will be able to engage confidently with such complexity. The graduate will recognise that there are no simple solutions to problems in society and that there are many twists and turns, dead ends and necessary restarts, and that they will need to act with resilience to succeed in these journeys. Such journeys will typically occur within entrepreneurial, innovation and investigative/research activities.

3. Relational capability

A CPUT graduate will be able to stand in the shoes of others in order to understand their needs, values and cultures so that what is being worked on can have optimal effects and/or the best chances of success. in so doing, a CPUT graduates will be able to act with understanding of others different from themselves, at both the interpersonal and inter-professional level. Furthermore, CPUT graduates will understand, learn with, and so be able to engage with others for the best possible solutions to work and societal problems. This capability is of advantage beyond the university and influences how graduates work with community groups or in local government, and relates to caring as well as effectiveness. it can also apply to working inter-professionally where, for example, a roads engineer would also have to work with environmentalists, heritage experts and others to get a job done for the benefit of all.

4. Ethical capability

CPUT graduates will be orientated towards the well-being and improvement of society rather than just ensuring the well-being and advancement of themselves. This will involve hearing and acknowledging the concerns of others. in the professional fields, furthermore, morality forms a cornerstone against which ethical decisions are made in practice and as such characterises being professional.